

Interpolation Attention-based KAN for the Sussex-Huawei Locomotion-Transportation Recognition Challenge

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Abstract

The Sussex-Huawei Locomotion-Transportation (SHL) recognition challenge organized at the HASCA Workshop of Ubicomp 2024 presents a large and realistic dataset with different activities and transportation. The goal of this human activity challenge is to recognize 8 modes of locomotion and transportation (activities) in a user-independent manner based on motion sensor data in an opportunistic manner. In this paper, our team (We can fly) summarize our submission to the competition. We proposed an interpolation attention-based Kolmogorov-Arnold network (KAN-IA), a deep learning method for missing variable representations recovery and transportation mode classification. Our KAN-IA model first extract individual motion sensor feature representations based on three KANs, then the missing and normal sensor representations outputs are fed into IA model to recover the missing variable representation. Finally, we use a linear layer to output the classification predictions. Our post-processing operations include three modules: PCA-based transformation, random masking of sensor data, and the ensemble learning. In the experiment, we achieved averaged F1 score of 0.55 on valid datasets.

Ccs Concepts

•Human-centered computing → Ubiquitous and mobile computing; •Computing methodologies → Artificial intelligence; •Information systems → Data mining.

Keywords

transportation mode detection; variable missing; data mining; Kolmogorov-Arnold networks; time series recovery

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1 Introduction

In recent years, the exponential growth of computational power and smart mobile devices has enabled the advent of intelligent sensing and ubiquitous computing. Smartphone-based Human Activity Recognition (HAR) holds extensive potential for application across various domains, including scene awareness, positioning and navigation systems, monitoring of physical activity levels, and personalized advertising, thereby attracting significant attention [1]. Transportation modes detection (TMD) as a branch of human behavior recognition has a great substantial influence on human life, such as traffic management, route or parking recommendations, and footprint analysis [2].

The SHL recognition challenge Dataset [3-7] contains multimodal locomotion and transportation data, which can be used in an activity recognition challenge. The goal of this year edition is to recognize 8 modes of locomotion and transportation (activities) in a user-independent manner based on motion sensor data in an opportunistic manner.

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Based on the experience with previous challenge [9–11], we report that despite the application potential of TMD research, the practical implementation of TMD in real-world environments still faces challenges. Firstly, it is challenging to train a model using smartphone data collected at a specific body position and then test the model using data collected from a new position. Furthermore, although HAR models may achieve high accuracy on randomly selected test data, their performance often significantly declines when tested with new users because various users perform the same activity in a different way [8]. HELP [9] introduced a multiple fusion model that, after extracting hand-crafted features from sensor data, feeds these features into both a deep learning model and a machine learning model for late fusion. HYU-CSE [10] employs a dense convolutional neural network (DenseNet) which receives separate inputs from various sensor data, fuses them within the network's architecture, and produces a final decision at the network's output layer. We-can-fly [11] has implemented a 1D DenseNet model that operates on multi-channel sensor data in an integrated manner. Finally, previous challenges have not yet explored the potential impact on the development of user-independent TMD models under conditions of severe sensor data missing. Accurate pattern recognition in the context of incomplete time series sensor data presents an even more formidable challenge.

In this paper, we proposed an interpolation attention-based Kolmogorov-Arnold Network [12] (KAN-IA) model for identify the transportation classification with multi-channel sensor data random missing manner. We extract time-frequency features from three types of motion sensors independently. Then we leverage the two-layer KAN model to capture the robust transportation representation, and the output features of KAN are fed into an interpolation attention module for recovering the miss sensor feature. Finally, we combined recovered feature and original features to a fully connection layer for the final classification. Furthermore, we employ the multi-model fusion to improve the performance.

2 Shl 2024 Dataset

The SHL recognition challenge dataset encompasses eight distinct locomotion and transportation modes. The 2024 SHL challenge utilized a subset of this dataset, posing a task for participating teams to address. The dataset was segmented into three distinct parts: training, validation, and testing, by the organizers. It consists of 59 days for training, 6 days for validation, and 28 days for testing, each derived from a continuous dataset using a non-overlapping, 5-second sliding window. Samples are comprised of 500 sensor values captured at a 100 Hz sampling rate.

The training dataset exclusively features raw sensor data from a single user (user1) across four phone locations (bag, hips, torso, hand). Conversely, the validation dataset integrates sensor data from two additional users (a mix of user 2 and user 3), also across the four specified phone locations.

The primary goal of the 2024 SHL challenge is to recognize the user activities based on data from the "test" user's phone, which is a composite of user 2 and user 3. Unlike the training data, the validation and test data may have one or more sensors with missing data, represented as zero values. The challenge involves recognizing transportation modes in manners of user-independent, phone positioning uncertain and random data missing, marking it as a more formidable challenge in comparison to previous challenges.

3 Pre-Processing

3.1 PCA based coordinate alignment

Due to the random motion sensor data loss manner in SHL2024 challenge (especially the accelerometer), we cannot transform sensor data from the device coordinate system to the navigation coordinate system using rotation matrix-based methods. Instead, we employed a principal component analysis (PCA)-based coordinate alignment method. This method first extracts the principal component axes of each sensor data frame, and then rotates the sensor data into the principal component coordinate system.

3.2 Adaptive Feature Extraction

Following the suggestion from previous SHL challenge, we extracted 1000-dimensional time-frequency features from the training dataset. Based on the performance of these features on the validation set, we adaptively selected 170 features, denoted as F_{sensor} , which primarily encompass statistical features, time-domain features, frequency-domain features, and time-frequency features.

Finally, we normalized the feature vectors using the minimum and maximum values of the feature vector. For the missing sensor axes, we imputed their missing feature data with zeros.

3.3 Random Masking

To enhance the model's ability to fit the expression of motion features when sensor data is randomly lost, during the training phase, we randomly mask one individual motion sensor data to simulate the scenario of validation set. This approach helps in mimicking the unpredictable nature of sensor data loss and trains the model to be robust against such variations.

4 Proposed Methods

KANs [12] have shown a more accuracy and interpretability representation performance in physics laws discovery compare with shadow deep neural networks, which have achieved success in previous challenge. Hence, we proposed a KAN-IA structure to extract the robust motion feature representation. Figure 1 shows the detail components of our KAN-IA structure, including input KAN block to extract transportation mode features. Then the outputs of the KAN layer will feed into the interpolation attention layer to recovery the mission sensor feature, the IA

layer contains a feature correspondences induction module and a mission sensor feature recovery module. The output of the IA layer will feed into the Liner layer to obtain the classification result.

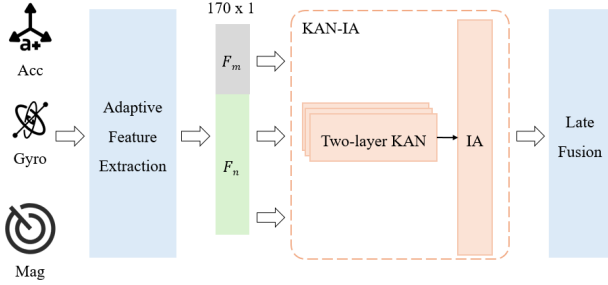


Figure 1: Proposed KAN-IA pipeline. The gray and green rectangles represent the randomly missing motion sensor features and the normal sensor features, respectively.

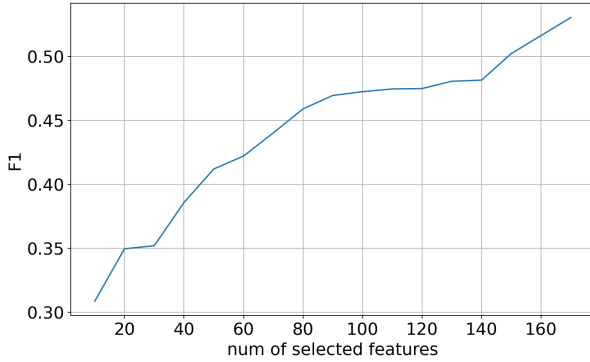


Figure 2: F1 score of the KAN-IA with different number of input feature size.

4.1 Input KANs Block

Inspired by the Kolmogorov-Arnold representation theorem, Kolmogorov Arnold Networks leverage spline-based learnable activation functions on network's weight parameter, which makes KANs outperform MLPs in terms of accuracy and interpretability. Our input KAN block consists of three two-layer KANs to extract features from our 170 dimensions adaptive features extracted by acceleration, gyroscope, and magnetometer measurements respectively. For each two-layer KAN, we define the number of KAN layer's unit as n and m , where n denotes the number of input sensor features (we set n as 64) and m denotes the axis number of output variable (we set m as 256). The KAN layer focus on extracts the relationship between each sensor features and generates three m feature-maps, we then concatenate the output of normal sensor features together to $2 * m$ feature-maps and feed it to IA layer with $1 * m$ missing sensor feature-maps.

4.2 Interpolation Attention Layer

Inspired by [13], we introduce the interpolation attention mechanism to recovery the mission sensor features. In the feature correspondences induction module, we first generate the corresponding metric between missing sensor features and normal sensor features as follows:

$$A_I = (I_N + \text{softmax}(\text{ReLU}(E_{A1}E_{A2}))) \quad (1)$$

Where $I_N \in R^{3 \times 3}$ is a diagonal matrix, $E_{A1} \in R^{3 \times m}$ and $E_{A2} \in R^{m \times 3}$ is two sensor feature embedding matrices, the value of row i and column j in the A_I denote the correlation between sensor type i and sensor type j . The attention coefficient α_{ij} between the missing sensor i and normal sensor j can be calculated as follows:

$$\alpha_{ij} = \frac{\exp(\text{ReLU}(FC(h_{\text{missing}})))}{\sum \exp(\text{ReLU}(FC(h_{\text{normal}})))} \quad (2)$$

Where h_{missing} and h_{normal} are the missing sensor output and normal sensor output of KAN layers, then we can recover the missing sensor representation according to the normal sensor representation as follows:

$$h_{\text{missing}}^i = \text{ReLU}\left(\sum_{n \in \text{Normal}} A_{in} \sum_{j=1}^m \alpha_{ij} \cdot h_{\text{normal}}^n\right) \quad (3)$$

After recovery, the outputs of the IA layer are fed into a Liner layer to obtain the classification result.

4.3 Late fusion

During the late fusion phase, we used our 170 dimension adaptive features as the training input and selected five base models, including XGBOOST, LIGHTGBM, DNN, ConvNext, and KAN (without IA), to predict classification results respectively, and combined them using a soft voting method as a type of ensemble strategy. Furthermore, before predicting activities in the test data, we fine-tuned the final trained models with the validation data.

5 Experiments

5.1 Computational Resources

We utilized one GPU on server.

- 1 GPU (Nvidia A800-SXM4-80GB), CPU 8-core@2.2GHz, RAM-200GB
- Programming Language: Python 3.11.0
- Framework: PyTorch 2.1

The size of the trained model is approximately 3.1MB. The model training took about one hour. The evaluation of the test dataset took approximately 2 minutes.

5.2 Experimental Results on Valid Datasets

Figure 2 illustrates the recognition performance of the KAN-IA model on the validation data with varying numbers of time-

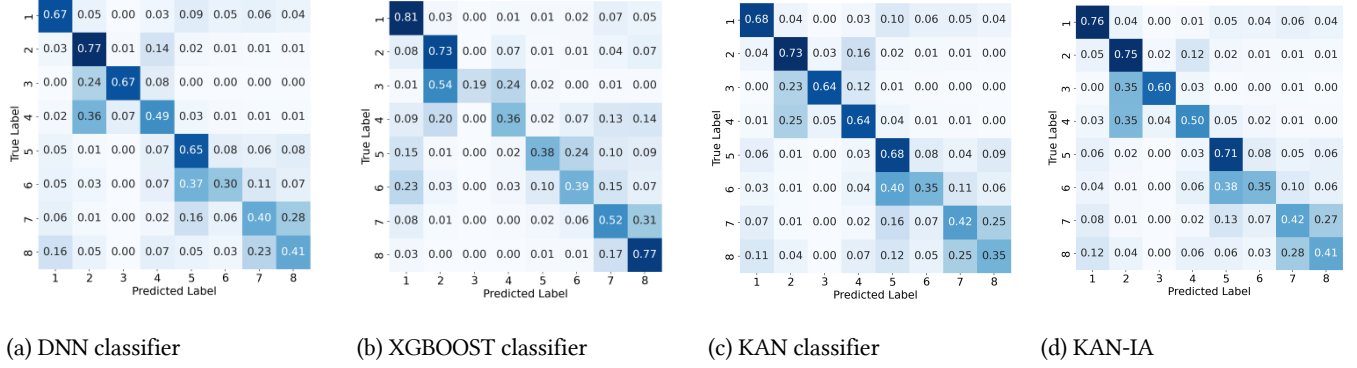


Figure 3: Confusion matrices of different model. The eight classes are: 1 - Still; 2 -Walk; 3 - Run; 4 - Bike; 5 - Car; 6 - Bus; 7 - Train; 8 - Subway

frequency features no longer appear to enhance model performance. Ultimately, by fine-tuning the selection of time-frequency feature vectors, we achieved an F1 score of 0.55.

Figures 3(a) and 3(b) present the prediction results from the DNN and XGBOOST models, which attained F1 scores of 0.51 and 0.53 on the validation set, respectively. It is observable that the DNN exhibits the highest predictive accuracy for category 2, while its precision for transportation-related activities ("Bus" and "Train") is lower. XGBOOST can effectively predict the "Still," "Walk," and "Subway" categories but struggles to differentiate between "Run" and "Bike." Figures 3(c) and 3(d) display the prediction outcomes of the KAN and KAN-IA models, which secured F1 scores of 0.50 and 0.55, respectively. According to the compare results, there is no marked difference in the accuracy rates of category predictions between the KAN and KAN-IA models, however, the KAN-IA model outperforms the KAN model in all categories except for "Bike" and "Run." Comparing the results in Figure 3(b) with other figures reveals that neural network-based models perform well in identifying categories 3, 4, 5, and 6, yet machine learning models exhibit superior performance in recognizing the "Subway" category.

6 Conclusion

In this study, we propose a user-independent Temporal Multimodal Data (TMD) model which can recover the random missing of sensor representations, using data from three smartphone sensors (accelerometer, gyroscope, and magnetometer, which are subject to substantial random sensor data missing) to detect eight transportation modes in the SHL recognition challenge. The TMD model is designed using the KAN architecture and Interpolation Attention (IA). The KAN architecture automatically extracts motion pattern

frequency feature vectors, measured by the F1 score. As the size of the model's input vector increases, the F1 score correspondingly increases and then stabilizes. Once the size of the feature vector exceeds 170, augmenting the quantity of time-

representations from the time-frequency features of the three smartphone sensors. Subsequently, the IA is employed for the recovery of missing sensors. Finally, a Linear layer is utilized as the classifier. The TMD model demonstrated an F1 score of 0.55 on the validation dataset. To enhance detection accuracy, we applied a soft voting mechanism of multiple models as post-processing. The recognition result for the testing dataset will be presented in the summary paper of the challenge [7].

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